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Integrating multi-task and cost-sensitive learning for predicting mortality risk of chronic diseases in the elderly using real-world data

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ABSTRACT

Background and Objective: Real-world data encompass population diversity, enabling insights into chronic disease mortality risk among the elderly. Deep learning excels on large datasets, offering promise for real-world data. However, current models focus on single diseases, neglecting comorbidities prevalent in patients. Moreover, mortality is infrequent compared to illness, causing extreme class imbalance that impedes reliable prediction. We aim to develop a deep learning framework that accurately forecasts mortality risk from real-world data by addressing comorbidities and class imbalance.

Methods: We integrated multi-task and cost-sensitive learning, developing an enhanced deep neural network architecture that extends multi-task learning to predict mortality risk across multiple chronic diseases. Each patient cohort with a chronic disease was assigned to a separate task, with shared lower-level parameters capturing inter-disease complexities through distinct top-level networks. Cost-sensitive functions were incorporated to ensure learning of positive class characteristics for each task and achieve accurate prediction of the risk of death from multiple chronic diseases.

Results: Our study covers 15 prevalent chronic diseases and is experimented with real-world data from 482,145 patients (including 9,516 deaths) in Shenzhen, China. The proposed model is compared with six models including three machine learning models: logistic regression, XGBoost, and CatBoost, and three state-of-the-art deep learning models: 1D-CNN, TabNet, and Saint. The experimental results show that, compared with the other compared algorithms, MTL-CSDNN has better prediction results on the test set (ACC=0.99, REC=0.99, PRAUC=0.97, MCC=0.98, G-means = 0.98).

Conclusions: Our method provides valuable insights into leveraging real-world data for precise multi-disease mortality risk prediction, offering potential applications in optimizing chronic disease management, enhancing well-being, and reducing healthcare costs for the elderly population.

1. Introduction

Chronic diseases are prevalent in the elderly population and are a major cause of their mortality [1,2]. For elderly people with multiple chronic diseases, the most appropriate treatment may depend on their life expectancy (i.e., their risk of death) [3]. Accurate prediction can provide early intervention opportunities and optimize treatment plans for different risk groups so that health resources can be allocated more rationally and the health of the overall population improved. It also provides a basis for the formulation of chronic disease management policies in a particular region [4,5].

Compared to controlled-cohort data, real-world data includes a wide range of individuals in the general population, and is, therefore, highly representative [6]. The use of real-world data in research can improve

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the generalizability of findings by taking into account factors such as patients' genetics, comorbidities, medication history, and treatment adherence [7]. Real-world data are further selected without an explicit selection bias, have high heterogeneity, and the results have high external validity. Forecasting using real-world data maximizes the insights provided to decision-making bodies, thereby enabling them to predict the burden of disease in a given region. Such foresight plays a crucial role in planning ahead and optimizing the allocation of healthcare resources [8–11].

The development of realistic risk prediction models requires the choice of appropriate machine learning (ML) methods, in particular, deep learning techniques that work well, even when employed on large datasets [12]. Current research focuses on predicting a single chronic disease or predicting the risk for a single patient population [13–15]. Comorbidities are prevalent in real-world data on chronic diseases [5,16–18]. However, so far, there are still a number of challenges affecting the prediction of mortality risk for multiple chronic diseases in real-world populations. This is attributed to the inherent complexities of modeling comorbidity, wherein heterogeneous comorbid patients exhibit distinct features and trajectories, especially with an increasing number of concurrent conditions. Moreover, diverse associations exist amongst chronic diseases, such as commonly observed comorbidity patterns in the elderly [19–22]. Hence, using a global deep model to accurately predict mortality risks across multiple chronic diseases is infeasible. In addition, in real-world data on chronic disease events, the rate of disease diagnosis far exceeds the incidence of mortality, even when the study population is elderly. Such highly class-imbalanced data pose further obstacles to reliable predictive learning [23–26]. Therefore, when confronted with real-world data, a novel deep learning framework that learns correlations and specificities of multiple chronic diseases and is also able to learn enough information about a few additional categories may improve the prediction of mortality risk for multiple chronic diseases.

In this study, we used real-world data on the elderly population from Shenzhen Healthy Aging Research (SHARE) to present an innovative deep learning-based framework for multiple chronic disease mortality risk prediction. First, we grouped patients with different chronic diseases based on indicative features, with each group representing patients with one chronic disease and one subtask. Then, we proposed the MTL-CSDNN model, which is an integrated multi-task [27,28], cost-sensitive [29] model. Multi-task deep learning has been widely used and progressed in the fields of medical image computation, analysis and disease prediction [30–34]. However, to date, no study has effectively applied multitask learning for mortality risk prediction of multiple chronic diseases, which differs from a global deep neural network by designing shared and task-specific layers in the hidden layer, thus giving the model the ability to encode common underlying patterns while maintaining the flexibility of personalized prediction by sharing lower embeddings and utilizing disease-specific higher layers. At the same time, we designed a cost-sensitive matrix and embedded it in the loss function to address the class imbalance problem. We attempted to capture the correlations and specificities of multiple chronic diseases through our model and demonstrate its usability through interpretable experiments. In particular, our aim was to accurately predict the risk of death from a large amount of data on elderly patients with chronic diseases in order to identify high-risk patients and provide a basis for the subsequent design of appropriate health control programs. Our study encompasses 15 prevalent chronic diseases, and the proposed AI methodology was trained and tested on real-world data from 500,000 elderly people aged 65 years old and older in Shenzhen, China. Each patient sample was used as a model input, and the mortality risks of patients in different chronic disease groups corresponded with different subtask outputs. The results show that our model has strong predictive power; it is reliable and outperforms other state-of-the-art artificial intelligence models. In summary, our main contributions are as follows.

First, we propose a new prediction goal using real-world data for multiple chronic disease mortality risk prediction in the elderly. To the best of our knowledge, this is the first study to predict the risk of multiple chronic disease mortality using data from a large cohort of elderly people with a wide range of data integrations, such as sociodemographic variables, lifestyle data, and physical examination data.

Second, this work pioneers the application of multi-task learning to extract insights across multiple chronic conditions and propose an MTL-CSDNN model to predict mortality risk.

Finally, through rigorous comparative assessments and ablations of a 500,000-patient Shenzhen cohort, we demonstrated that MTL-CSDNN attains stellar performance, with all metrics surpassing 0.97. This significantly outperforms baseline models, validating its capabilities for large-scale multimorbidity mortality risk screening and providing a vital decision-support tool for chronic disease management.

2. Methods

2.1. Data

The real-world data used in this study relied on the older adult health management project of the National Basic Public Health Service in Shenzhen that began in 2018, which recruited people aged 65 and older from the lists of all residents registered at local community health centres. Since 2018, new recruitment and follow-up surveys have taken place almost every year, with each wave taking 12 months to complete. Details of the recruiting and survey procedure were previously published elsewhere [35]. We maximized the use of all available data, encompassing participants surveyed from January 1, 2018, to December 31, 2021, to constitute the study cohort. A total of 937,064 data points were acquired, of which 502,688 participants were enrolled (the number of elderly people aged 65 years old and older in the data of the Seventh Population Census of Shenzhen was 565,217), and we selected the most recent data for each participant as our metadata. Of the participants, 225,538 were men, with a median age of 69 years (interquartile (IQR): 67–73 years), and 277,150 were women, with a median age of 69 years (interquartile (IQR): 66–73 years). The vital status of the participants by December 31, 2021, was determined in 3 steps, including linkage with the China Population Death Information Registration System, the detailed process have been described in previous publications [36,37]. Our study was approved by the Shenzhen Center for Chronic Disease Control Human Ethics Committee (No. SZCCC-2021–061-01-PJ). As claims data were provided anonymously, this study was exempted from the requirement for informed consent by the Shenzhen Center for Chronic Disease Control Human Ethics Committee.

This real-world cohort contains extensive multidimensional data. The standardized questionnaires assessed sociodemographic characteristics, lifestyle factors and health parameters, which together provide information regarding the health statuses and risks of the participants. Meanwhile, physical examination, electrophysiological examination and biochemical analyses, provided objective clinical assessments. The full details of the survey procedures, measurement methods, and baseline variables have been described in previous publications [35,38,39]. All 64 features included in this study are shown in the supplementary material. These heterogeneous inputs provided multifaceted insights into a patient's physical condition, and together formed a large-scale tabular dataset, with each row representing a participant record and the columns capturing a range of attributes (Fig. 1a).

2.2. Chronic diseases subgroup

The aim of this study was to predict the mortality risk in the elderly with chronic diseases, and we distinguished what diseases the subjects had, if any, based on their characteristics, e.g., hyperuricemia was defined as a serum uric acid level of > 420umol/L [40], and all diseases and conditions were defined as dichotomous variables (present vs. absent). After combining the number of patients with each chronic disease

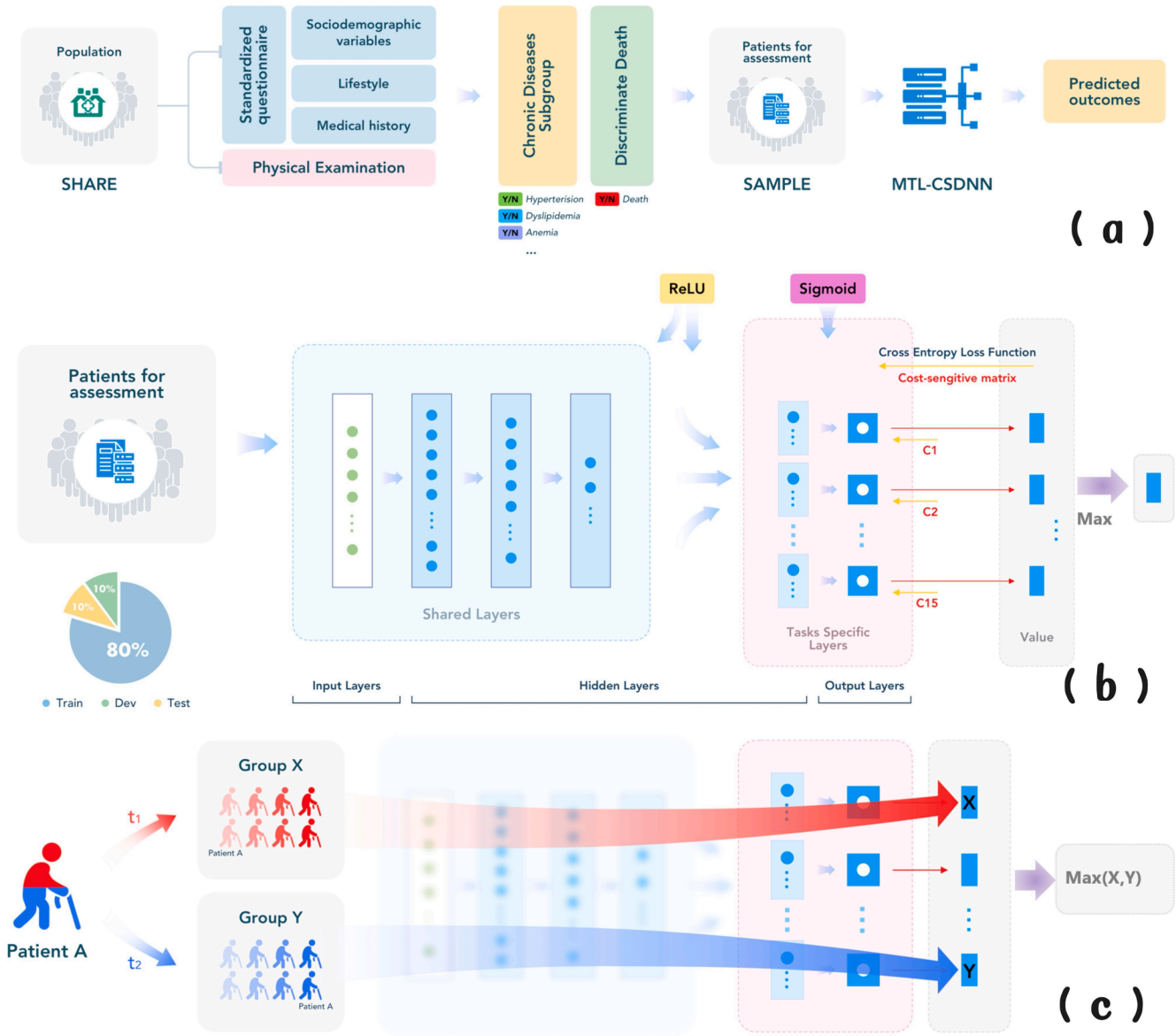

Fig. 1. Framework for predicting mortality risk of multiple chronic diseases in a real elderly population. a, Overview: the study describes collecting data from the general population through standard questionnaires, physical examinations, etc., then grouping chronic diseases and determining mortality to obtain samples as input to our model, and finally outputting mortality risk. b, Model architecture: MTL-CSDNN leverages shared low-level representations and task-specific top layers to enable accurate prediction across multiple chronic diseases. In line with standard deep learning workflows, we split the real-world dataset into training, validation, and test sets. The training set is used to update model parameters, the validation set provides unbiased evaluation during training for hyperparameter tuning, and the final test set evaluates the model's generalization on new data. c, Learning: a microscopic description of the model learning one sample. The patient is divided into multiple chronic disease groups before entering the model. t1, t2 indicate this is an asynchronous learning process, corresponding to different subtask outputs.

and the core list of chronic diseases recommended by the systematic evaluation [41], 15 high-prevalence chronic diseases were finally screened. Data from a total of 482,145 patients were obtained, including 217,584 males and 264,561 females, with a disease prevalence of 95.91 % and a total of 9,516 mortality events (1.89 %).

To enable multi-task learning, patients were assigned to distinct subgroups based on their disease profiles, with multimorbid individuals being allocated to multiple groups accordingly. For example, a patient with both dyslipidemia and diabetes was assigned to both subgroups. Each disease-defined subgroup represented a separate prediction task, with related multimorbid tasks allowing for knowledge-sharing during modeling. We delineated the inclusion criteria, cohort sizes, comorbidity proportions, and mortality counts for each chronic disease task in Table 1. This principled data structuring provides an ideal testbed for multi-task learning that is aligned with clinical realities while retaining the original data distributions.

2.3. MTL-CSDNN architecture

The MTL-CSDNN framework employs a multi-task learning architecture for deep neural networks (Fig. 1b), deviating from conventional monolithic DNNs. It comprises an input layer, an output layer, and four hidden layers. A hard parameter-sharing approach is utilized [28], with the input and the first three hidden layers shared across all tasks, while the fourth hidden layer is task-specific and linked to the output layer of each corresponding task. Here, each task represents one chronic disease group. In the depicted multi-task deep neural network learning framework, each sample is pre-assigned to distinct tasks before the input layer is entered. As shown in the Fig. 1c, a patient afflicted with diseases X and Y would be fed into the model twice, once for task X and again for task Y,

Table 1
Datasets: 15 chronic diseases involved in the study, number of patients with each chronic disease, multimorbidity and mortality.

Chronic diseases (subtasks)	Number	Multimorbidity	Mortality	Chronic diseases (subtasks)	Number	Multimorbidity	Mortality
Dyslipidemia (T1)	269885 (53.69 %)	258733 (95.87 %)	4773	Hyperuricemia (T9)	92898 (18.48 %)	91404 (98.39 %)	785
Hearing lmpairment (T2)	26253 (5.22 %)	25696 (97.88 %)	1125	Chronic Kidney Disease (T10)	43371 (8.63 %)	42904 (98.92 %)	1939
Anemia (T3)	93103 (18.52 %)	87312 (93.78 %)	3735	Renal Calculi (T11)	57903 (11.52 %)	56460 (97.51 %)	340
Cardiovascular Disease (T4)	71763 (14.28 %)	70569 (98.34 %)	1840	Renal Cyst (T12)	87085 (17.32 %)	84984 (97.59 %)	646
Cerebrovascular Disease (T5)	30265 (6.02 %)	30011 (99.16 %)	968	Hepatic Insufficiency (T13)	168071 (33.43 %)	160294 (95.37 %)	2801
Diabetes Mellitus (T6)	140068 (27.86 %)	137645 (98.27 %)	3338	Hepatic Cyst (T14)	72423 (14.41 %)	69922 (96.55 %)	1005
Cholecystolithiasis (T7)	46837 (9.32 %)	45854 (97.90 %)	1002	Fatty Liver (T15)	191476 (38.09 %)	187156 (97.74 %)	2134
Hypertension (T8)	334451 (66.53 %)	316868 (94.74 %)	6493				

thus eliciting outputs from both tasks X and Y in an asynchronous process. The prediction results of each subtask reflect the severity of each disease to a certain extent. Considering that taking the average value may mask certain diseases with very high severity, from the perspective of clinical interpretability, we chose the maximum value of each subtask as the final risk of death for a single patient, which we believe may better reflect the actual risk profile of an individual. This enables the model to learn specialized representations and make personalized predictions for patients with multiple comorbid conditions. By partitioning the data into subtasks based on each disease profile prior to modeling, the algorithm can explicitly capture both shared and task-specific patterns underlying the heterogeneity of real-world data. Such an approach circumvents the limitations of conventional single-task models that fail to account for comorbidity correlations.

The large-scale chronic disease cohort presents an extreme class imbalance, with relatively few mortality events (1–4 %) in each disease group. This skewed distribution poses obstacles for standard classifiers, which often neglect rare outcomes. To address this issue, we formulated the binary mortality prediction as a cost-sensitive learning task [29]. As depicted, we integrated cost sensitivity into the MTL-CSDNN architecture by embedding a cost-sensitive matrix into the loss function. During backpropagation, this matrix penalizes classification errors in a directed asymmetric manner, with expenses scaled to reflect the severity of different misclassification types. By propagating these weighted penalties back through the network, the model is incentivized to prioritize the correct identification of mortality events to minimize overall costs. This principled augmentation of standard loss formulations with domain-specific misclassification expenses enables a tailored optimization of medical classification tasks wherein all errors are not equal (Fig. 1b).

2.4. MTL-CSDNN algorithm

At an algorithmic level, our proposed multitask learning framework is formalized as follows:

For neuron $\mathbf{j}$ present in layer $\mathbf{l}$, let the weight vector $\mathbf{w}^{\mathbf{l}}_{\mathbf{jk}}$ denote the weight of the $\mathbf{k}$ neuron in layer $\mathbf{l}-1$ associated with the jth neuron in layer $\mathbf{l}$:

$$\mathbf{a}^{\mathbf{l}}_{\mathbf{j}} = \sigma\left(\mathbf{z}^{\mathbf{l}}_{\mathbf{j}}\right) = \sigma\left(\sum_{\mathbf{k}=1}^{\mathbf{m}} \mathbf{w}^{\mathbf{l}}_{\mathbf{jk}} \mathbf{a}^{\mathbf{l}-1}_{\mathbf{k}} + \mathbf{b}^{\mathbf{l}}_{\mathbf{j}}\right) \mathbf{l} = 2, 3, 4$$

where $\mathbf{a}^{\mathbf{l}}_{\mathbf{j}}$ denotes the output of the $\mathbf{j}$ neuron in layer $\mathbf{l}$, $\sigma()$ is the activation function, $\sigma() = \mathbf{ReLU}()$, $\mathbf{b}$ is the bias, and we default to the number of neurons in the previous layer denoted as $\mathbf{m}$. When $\mathbf{l} = 2$, $\mathbf{x_k}$ denotes the $\mathbf{k}$ initial neuron.

We perform MTL framework construction in the fourth hidden layer, for the $\mathbf{i}$ task there:

$$\mathbf{a}^{\mathbf{l_i}}_{\mathbf{j}} = \sigma\left(\mathbf{z}^{\mathbf{l_i}}_{\mathbf{j}}\right) = \sigma\left(\sum_{\mathbf{k}=1}^{\mathbf{m}} \mathbf{w}^{\mathbf{l_i}}_{\mathbf{jk}} \mathbf{a}^{\mathbf{l_i}-1}_{\mathbf{k}} + \mathbf{b}^{\mathbf{l_i}}_{\mathbf{j}}\right) \mathbf{l} = 5$$

Each task corresponds to an output:

$$\mathbf{y_i} = \mathbf{Sigmoid}\left(\sum_{\mathbf{k}=1}^{\mathbf{m}} \mathbf{w}^{\mathbf{l_i}}_{\mathbf{k}} \mathbf{a}^{\mathbf{l_i}-1}_{\mathbf{k}} + \mathbf{b}^{\mathbf{l_i}}\right) \mathbf{l} = 6$$

The cost matrix encodes the penalty for classifying a sample from one category to another. We denote by $\mathbf{C_i}(\mathbf{p}, \mathbf{q})$ the cost of predicting a sample of class p to a sample of class $\mathbf{q}$ in the ith task, and $\mathbf{p}$ is the total number of classes. Our problem is a binary classification problem, so $\mathbf{p} = 2$. The diagonal elements in the cost sensitivity matrix correspond to the category weights, respectively, and all other elements are 0. Larger values in the cost sensitivity matrix result in larger penalties. In our experiments, we take values in the cost matrix based on a parametric learning method. In this work, we use a modified cross-entropy loss function, which is embedded in the cost-sensitive matrix as follows:

$$\varepsilon_{\mathbf{i}} = -\sum_{\mathbf{t}} \mathbf{C^t_i}(\mathbf{p}, \mathbf{q}) \mathbf{Y^t_i} \mathbf{log}(\mathbf{y^t_i})$$

where $\varepsilon_{\mathbf{i}}$ denotes the loss function for the $\mathbf{i}$ task and $\mathbf{Y^t_i}$ denotes the true value of the $\mathbf{t}$ sample in task $\mathbf{i}$.

The loss function of the shared hidden layer is:

$$\varepsilon = \frac{1}{\mathbf{i}} \sum_{\mathbf{i}} \varepsilon_{\mathbf{i}}$$

2.5. Experimental setup

2.5.1. Baseline model

Mortality risk prediction across multiple chronic diseases using real-world data represents an unexplored task without previously established benchmarks. Therefore, we shift perspectives to AI advancements specialized for tabular data. While deep learning models are considered to excel at extracting complex data relationships and providing accurate predictions [42], it is impossible to ignore the finding of recent studies that tree-based methods demonstrate improved state-of-the-art performance over deep learning on many tabular data [43,44]. Hence, our comparative assessment utilizes both paradigms.

In machine learning, logistic regression models are commonly used in the medical field since they can handle binary classification problems well and do not have high requirements regarding the distribution of data types [45]. Tree integration models, such as XGBoost [46] and CatBoost [47], are considered the recommended choice for real-life tabular data problems as they achieve state-of-the-art results on many

tabular data. Among the deep learning models that have been recently proposed for learning from tabular data, we have selected three that are considered superior to tree models. First, TabNet [48] is a new high-performance, interpretable canonical deep tabular data learning architecture that allows for end-to-end learning, which is in line with our work, and it performs well on multiple datasets. Similar to the TabNet model, the SAINT [49] model is also based on transformers, with the added advantage that when a feature in a given row is missing or noisy, the inter-sample attention allows it to borrow the corresponding feature from other, similar data samples in the batch, meaning that it theoretically excels at learning from real-world data. Finally, the utilization of tabular data in Kaggle competitions was exemplified by the application of 1D-CNN [50]. Notably, the competition witnessed exceptional single-model performance, attaining optimal results through adept feature extraction facilitated by a convolutional neural network (CNN) architecture.

2.5.2. Data partitioning and model optimization strategies

To impartially evaluate predictive capabilities on real-world data, we partitioned the cohort into train (385716), validation (48214), and test (48215) sets with an 8:1:1 ratio. All baselines utilized the same splits to allow for fair comparisons. Initial experiments revealed an extreme class imbalance—mortality events were heavily outnumbered by chronic disease patients. This manifested in an inflated validation accuracy above 95 % but near-zero recall for all baselines as models neglected the rare positive class. To enable more rigorous assessments, we refined the baselines with techniques suited to their model family. As decision trees do not directly optimize a loss, oversampling was used to augment minority samples prior to training. For logistic regression and neural networks, cost-sensitive learning provided an elegant way to embed class weighting into the loss function. Together, these techniques ameliorated skewness and facilitated learning from scarce mortality data. The improved baselines now offer challenging yet fair comparisons to analyze the capabilities of our proposed approach on this real-world risk prediction task with calibrated predictive performance.

We provide comprehensive details regarding the search spaces and optimal hyperparameters selected for each model, fully disclosing the precise experimental settings used to attain the reported results. The key hyperparameters for MTL-CSDNN are set as follows: training epochs of 60, positive class weight (pos weight) of 30, negative class weight (neg weight) of 10, learning rate of 0.01, and a batch size of 256. The optimization was performed using Stochastic Gradient Descent (SGD) with a momentum of 0.9 and dampening of 0.1. Additionally, a learning rate decay strategy was applied with a step size of 4 and a decay factor (gamma) of 0.8. The logistic regression model parameters are the same as for MTL-CSDNN. For other baseline models, Bayesian hyperparameter optimization was conducted, selecting the minimum-loss configuration on the validation split for final evaluation. This principled approach ensures unbiased model selection and performance reporting. All models utilized early stopping on the validation data as regularization. Such methodological transparency is essential for imparting trust and facilitating the reproducibility of studies that utilize AI methods for risk prediction of mortality from multiple chronic diseases in the elderly on the basis of real-world data. The list of hyperparameters, search spaces, and optimal parameter combinations for all baseline models in this paper are in the supplementary file.

2.5.3. Evaluation metrics

We evaluated model performance regarding mortality risk prediction across multiple chronic diseases in the elderly using five metrics: accuracy (ACC), recall (REC), precision-recall AUC (PR-AUC), Matthews correlation coefficient (MCC), and G-means. Accuracy measures correct classifications overall, thereby assessing generalizability. Recall quantifies the true positive rate, which was chosen to evaluate model sensitivity in terms of capturing rare mortality events. The PR-AUC summarizes the trade-off between precision and recall, thus providing a robust metric for imbalanced data. The MCC calculates the correlation between predictions and ground truth, indicating predictive reliability. The G-means calculate the geometric mean of recall for each class, thus evaluating balanced performance across classes. This comprehensive suite of metrics provides a holistic profile of performance spanning the overall accuracy, identification of rare outcomes, precision-recall balance, predictive consistency, and balanced class-wise sensitivity. Taken together, these provide a rigorous evidence-based assessment of the real-world viability of the proposed approach for heterogeneous multiple morbidity mortality risk stratification.

3. Results

3.1. Evaluation of model performance

To effectively gauge the performance of our proposed model, this section commences with an analysis of its accuracy in predicting individual chronic conditions, treated as distinct subtasks within our framework. Initially, we evaluate the predictive capabilities of the MTL-CSDNN model for each subtask, utilizing five key performance metrics. These indicators were derived from the outcomes on a strictly withheld test set, thereby ensuring that the results remain completely independent from influences of the training process and hyperparameter tuning. The results, as shown in Table 2, indicate exceptional performance across all 15 subtasks. Specifically, the model achieves its highest accuracy in the subtasks of hyperuricemia and renal calculi, reaching 99.99 %, while it shows relatively weaker performance in the dyslipidemia subgroup with an accuracy of 99.15 %. In terms of recall, the renal cyst subtask excels with a recall rate of 99.99 %, whereas the hearing impairment subtask lags behind with the lowest recall of 97.53 %. The renal calculi subtask also leads in PR-AUC performance at 99.43 %, with the hypertension subtask performing the poorest at 93.17 %. As for the Matthews correlation coefficient (MCC) and G-means, the best-performing subtasks include hearing impairment, cerebrovascular disease, and renal calculi, all achieving 99.99 %. In contrast, the hypertension subtask records the lowest MCC and G-means scores at 91.11 % and 91.30 %, respectively. Despite the variability in model performance across different chronic diseases, these discrepancies are relatively minor, with no subtask deviating by more than 9 %. The lowest values across all five evaluation metrics never fall below 91 %.

We further comprehensively evaluated the predictive capabilities of different models on real-world data using five key metrics. It is worth noting that the overall performance metrics for the MTL-CSDNN model are based on averaging the performance in each subtask. All performance metrics were computed by applying each trained risk assessment model to a strictly withheld test set, thus ensuring its independence during both the training and hyperparameter search. In the final performance assessment of the test set for various types of AI models, MTL-CSDNN demonstrated exceptional results (Table 3); its ACC reached 0.9985, surpassing the second-ranked XGBoost (0.9384) by 6 percentage points. Additionally, our model achieved a REC of 0.9954, a 20-percentage-point improvement over TabNet (0.7925). More importantly, the PR-AUC (0.9722) and MCC (0.9790) of our model significantly outperformed all baseline models. The G-means value for MTL-CSDNN was 0.9795, outperforming the second-ranked TabNet (0.7965) by 18 percentage points. These outstanding performance metrics highlight the superior predictive capabilities of MTL-CSDNN on real-world data. Specifically, not only does it outperform competitors in accuracy and recall but it also exhibits a clear advantage in terms of key metrics, such as PR-AUC and MCC. Thus, MTL-CSDNN provides a powerful and reliable solution for the risk prediction of mortality from multiple chronic diseases in elderly populations.

3.2. Ablation study

To comprehend and assess the contributions of multitask learning

Table 2
MTL-CSDNN performance in 15 subtasks.

	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15
ACC	**0.9915**	0.9926	0.9976	0.9998	0.9941	0.9934	0.9949	0.9964	**0.9999**	0.9996	**0.9999**	0.9949	0.9982	0.9997	0.9991
REC	0.9812	**0.9753**	0.9812	0.9998	0.9956	0.9977	0.9973	0.9819	0.9994	0.9994	0.9998	**0.9999**	0.9989	0.9997	0.9995
PR-AUC	0.9458	0.9562	0.9570	0.9752	0.9661	0.9727	0.9790	**0.9317**	0.9920	0.9555	**0.9943**	0.9931	0.9724	0.9854	0.9739
MCC	0.9328	**0.9999**	0.9697	0.9951	**0.9999**	0.9672	0.9978	**0.9111**	0.9981	0.9953	**0.9999**	0.9989	0.9465	0.9816	0.9629
G-means	0.9340	**0.9999**	0.9709	0.9952	**0.9999**	0.9681	0.9979	**0.9130**	0.9981	0.9955	**0.9999**	0.9990	0.9471	0.9847	0.9633

Table 3
Performance of mortality risk prediction models.

		ACC	REC	PR-AUC	MCC	G-means
ML	LR	0.8798	0.6610	0.1915	0.2216	0.2537
	XGBoost	0.9384	0.5317	0.3343	0.2742	0.2980
	CatBoost	0.8855	0.7080	0.3941	0.2531	0.2840
DL	1D-CNN	0.9167	0.5957	0.2055	0.2487	0.2756
	TabNet	0.7999	0.7925	0.3538	0.5998	0.7965
	Saint	0.9164	0.5658	0.1804	0.2355	0.2630
	MTL-CSDNN	**0.9985**	**0.9954**	**0.9722**	**0.9790**	**0.9795**

and cost sensitivity to the overall performance of the model, we conducted further ablation experiments focusing on key components of the model. These experiments aim to address the following questions: Does the multitask learning framework play a decisive role? Can a global DNN achieve comparable experimental outcomes? Is it necessary to incorporate cost sensitivity into the model? To address these inquiries, three variant models were designed, namely DNN, CSDNN, and MTL-DNN. Architecturally identical to MTL-CSDNN, these variants isolate the incremental benefits of multitask knowledge-sharing and asymmetric misclassification costs.

Table 4 presents a comprehensive comparison of the original model, MTL-CSDNN, against the three variant models across various performance metrics. Although DNN achieves an ACC of 0.9814, the REC is 0, which is similar to the results demonstrated in our earlier experiments with the baseline model, again ignoring the rare positive class. We added cost sensitivity to DNN, and the REC increased, but only to 0.3578, while the ACC decreased to 0.9038, and the values of the other metrics also remained very low. The superiority of cost sensitivity validates its effectiveness in guiding the model to be sensitive to the identification of a critical minority of cases, but the model is still unable to capture the relevant features well enough to support the prediction task. The MTL-DNN model performance shows substantial improvement, with a REC of 0.8510 and ACC (0.9960), PR-AUC (0.7472), and MCC (0.7320) values also being higher than the experimental results of any baseline model, and the G-means (0.7366) value being only lower than that of TabNet (0.7965), which confirms that multitask learning successfully captures subtle disease associations hidden in the global model. However, it still falls short of the MTL-CSDNN proposed in this paper, which suggests the necessity of adding cost-sensitive learning.

3.3. Model explanation

3.3.1. Local explanation

In our study, we utilized Shapley Additive Explanations (SHAP) values to conduct a detailed feature importance analysis for each subtask within MTL-CSDN. This approach allows us to dissect and understand the influence of individual predictors on the model's outcomes. Across the fifteen subtasks addressed, SHAP plots were generated to visually represent the contributions of the top 20 most impactful features for each condition (Fig. 2).

For the purpose of detailed exposition in this manuscript, we selected two subtasks for in-depth analysis, while the SHAP plots for the remaining thirteen subtasks are provided in the supplementary materials. Within the chronic kidney disease subtask, the analysis revealed that serum creatinine, lower limb edema 2, and blood urea levels are the most significant predictors, all demonstrating a positive correlation with mortality outcomes. These findings suggest that elevated levels of these biomarkers are strongly indicative of higher mortality risks, emphasizing their pivotal roles in clinical assessments and decision-making processes for patients with renal impairments. Similarly, in the subtask focusing on hepatic insufficiency, the SHAP values identified lower limb edema 2, age, and total bilirubin levels as the three key contributors to mortality outcomes. This pattern indicates that older patients with higher levels of bilirubin and those exhibiting lower limb edema are at a significantly increased risk of mortality, underlining the necessity of vigilant monitoring and management strategies for this patient group.

Table 4
Comparison results of ablation experiments.

	ACC	REC	PR-AUC	MCC	G-means
MTL-CSDNN	**0.9985**	**0.9954**	**0.9722**	**0.9790**	**0.9795**
MTL-DNN	0.9960	0.8510	0.7472	0.7320	0.7366
CSDNN	0.9038	0.3578	0.1762	0.2205	0.2503
DNN	0.9814	0	0.1937	0	0

3.3.2. Task similarity measurement experiments

MTL-CSDNN achieves correlation modeling by mining correlation features between different tasks and sharing parameters at the bottom to extract shared features that can serve all tasks. To assess the effectiveness of the task similarity measurement in MTL-CSDNN, an analysis of shared parameters among different subtasks was conducted. If the model successfully learned the inter-task correlations, shared parameters would exhibit a certain degree of similarity across various subtasks. Building on this assumption, we employed a parameter similarity measurement method, specifically, cosine similarity, to quantify the similarities between parameters, thereby obtaining a correlation matrix depicting the inter-task relationships within the model.

To validate the obtained correlation matrix's real-world relevance, we statistically compiled the co-occurrence of patients with different chronic diseases in real-world data, constructing various chronic disease correlation matrices. By objectively reflecting the correlations between each pair of chronic diseases in the number of individuals concurrently afflicted, we conducted a similarity measurement between these two matrices. The correlation coefficient of 0.8033 was calculated by cosine similarity, it demonstrates the model's capacity to effectively capture any correlations between multiple chronic diseases, which is well-aligned with empirical realities and provides robust support for predictive tasks.

4. Discussion

We proposed MTL-CSDNN, a DL model applied to real-world data to predict mortality risks across multiple chronic diseases in the elderly. In particular, this study aimed to leverage a dataset of 500,000 adults aged 65 years old and older with chronic conditions, living in Shenzhen, China, and evaluate the model's mortality risk prediction capabilities through added baselines and ablation experiments. Our results show that MTL-CSDNN can capture complex relationships between multiple chronic diseases and handle class imbalance well, achieving high-

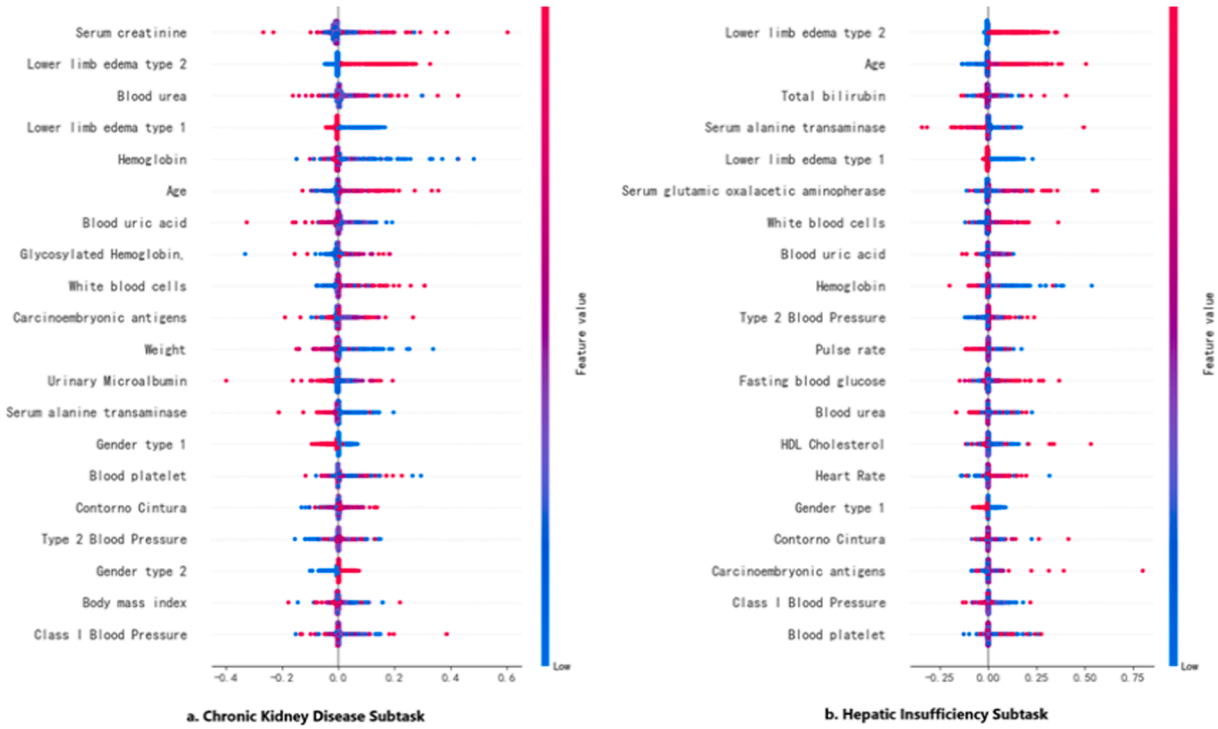

Fig. 2. The SHAP plots for the MTL-CSDNN-based mortality risk prediction for multiple chronic diseases (a. Chronic Kidney Disease subtask; b. Hepatic Insufficiency subtask).

performance predictive capabilities.

The real-world cohort we utilized captures extensive participant characteristics via standardized questionnaires and clinical examinations, composing a large-scale tabular data resource ideal for our study. Its scale and diversity surpass constrained experimental samples, thus enhancing the external validity that is critical for generalizable risk models. Based on this dataset, we designed a prediction process with potentially significant real-world relevance. We grouped patients by chronic disease, facilitated multitask learning by preprocessing samples into disease-specific tasks, and primed the model with some prior knowledge at very low cost to improve predictions across multiple chronic diseases on this basis.

Comprehensive comparative experiments and ablation studies showed our model's superior performance over other models. Specifically, other models sacrificed accuracy for recall, including advanced transformer-based models, like TabNet (ACC=0.7999, REC=0.7925) and Saint (ACC=0.9164, REC=0.5658). MTL-CSDNN achieved both high accuracy (ACC=0.99) and recall (REC=0.99), indicating effective learning of complex disease relationships and minority class features. Our model achieved a PR-AUC of 0.98, suggesting high stability, which is critical for sensitive tasks, like mortality prediction in healthcare. The MCC of 0.9790 and G-means of 0.9795 further affirm MTL-CSDNN's consistent performance in mortality risk assessment for elderly individuals with multiple chronic diseases. These metrics, acknowledged in DL evaluation, highlight MTL-CSDNN's accuracy, reliability, balanced sensitivity, and specificity, which are crucial for effective risk prediction based on real-world data. Ablation studies demonstrated that multitask learning and cost-sensitivity are indispensable, with multitask learning playing a central role. Additionally, we performed interpretability experiments on the model. At the micro level, we utilized Shapley Additive Explanations (SHAP) values to conduct a detailed feature importance analysis for 15 subtasks within MTL-CSDN. This approach allows us to dissect and understand the influence of individual predictors on the model's outcomes. At the macro level, given the uniqueness of this study, to demonstrate that MTL-CSDNN is consistent with objective facts in terms of accurately capturing complex chronic disease relationships, we designed a novel task similarity measurement experiment, building a similarity matrix from shared parameter distances between tasks, and a chronic disease comorbidity matrix from real comorbid patients, before measuring the similarity between the two matrices to validate our model's accuracy.

In summary, our research findings have multiple implications for public health and chronic disease management. First, existing models predicting chronic diseases or related outcomes tend to focus on single diseases while real-world patients, especially the elderly, often have multiple concurrent chronic conditions. A model predicting mortality risk across multiple chronic diseases could, thus, inform the design of improved management and treatment plans. In addition, developing separate prediction models for each chronic disease is rather inefficient and costly, and such models may underperform by ignoring disease correlations.

The particular advantages of this study are as follows. First, computational screening of large populations can be accomplished at a low cost, while complex clinical screening and interventions are quite costly and limited to fewer high-risk patients. We only need to group the elderly by chronic disease using simple criteria before inputting the data into MTL-CSDNN without extensive feature engineering to obtain accurate predictions for all elderly chronic disease patients in a given region, thereby providing reliable support for stratified management. Second, compared to other models, our model has a simple structure, a low computational cost, few parameters, little need for prior knowledge, and is easy to understand and extend. Its real-world prediction performance may already be sufficient to provide some decision support in chronic disease management regionally, with great potential for further improvement.

This study also has several limitations. Firstly, our dataset is derived solely from the Shenzhen Healthy Aging Research, introducing a geographical bias that may limit the generalizability of our findings. Additionally, our model currently accommodates 15 prevalent chronic diseases and was not designed to be inherently adaptive. Thus, the integration of additional chronic disease types in the future might adversely affect the model's performance. Furthermore, although our

study design incorporates 64 feature variables, including sociodemographic variables, lifestyle data, and physical examination items, it did not include textual data, such as medical records, and did not account for variables containing a large number of missing values, such as education level, despite the possible benefits of these variables for the study.

In the future, with more comprehensive data collection, including unstructured data, and an increased need to predict mortality across more chronic diseases, an extensible prediction framework could emerge, covering more disease types, enabling large-scale mortality risk prediction across unstructured data for broader patient populations, and improving both quality of life and the cost-effectiveness of chronic disease management. In subsequent research, we may focus on applying the model's knowledge for the stratified management of elderly chronic disease groups, conducting multifaceted analyses from the perspectives of improving patients' quality of life, life expectancy, cost-reduction, etc., and long-term management, tracking, and follow-up.

Code availability

Our repository is publicly available on https://github.com/QZQ-go/MTL-CSDNN. Experiments were conducted using Python version 3.9, torch version 2.0.1.

CRediT authorship contribution statement

Aosheng Cheng: Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yan Zhang:** Writing – original draft, Resources, Project administration, Investigation, Formal analysis. **Zhiqiang Qian:** Validation, Software, Project administration, Methodology. **Xueli Yuan:** Supervision, Resources, Project administration, Formal analysis, Data curation, Conceptualization. **Sumei Yao:** Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Wenqing Ni:** Writing – review & editing, Supervision, Resources. **Yijin Zheng:** Resources, Methodology, Investigation, Conceptualization. **Hongmin Zhang:** Resources, Conceptualization. **Quan Lu:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Zhiguang Zhao:** Writing – review & editing, Supervision, Resources, Project administration, Investigation, Funding acquisition, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The SHARE data that support the findings of this study are available on request from the corresponding author (Z.G.Zhao). The data are not publicly available due to them containing information that could compromise research participant privacy or consent.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijmedinf.2024.105567.

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